

Optimal dynamic portfolio selection for projects under a competence development model

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Abstract A model for optimizing investments of human resources over time into a given set of project classes with known competence requirements and known returns is presented. The model considers increments of competencies by learning as well as decrements by knowledge depreciation. We derive analytical results addressing questions such as whether an optimal policy will specialize on a single project class or rather keep a mixed project portfolio. In absence of uncertainty, gradual specialization turns out as advisable, but mixed portfolios in the intermediate term can be better than instantaneous specialization. Algorithms for optimizing the project portfolio selection are outlined.

Keywords Project portfolio selection · Learning · Knowledge depreciation · Resource allocation · Optimization

1 Introduction

Whereas the optimal selection of projects has been a traditional topic of investigation in the Operations Research literature [see, e.g., [Asher \(1962\)](#), [Hess \(1962\)](#) and [Bege-Dov \(1965\)](#) for early works], it was only recently that authors have begun to study a particular aspect of project selection in the context of formal optimization models: the mechanism of *competence development*, consisting of *learning* and *forgetting* effects related to competencies. By carrying out projects of certain types, competencies or skills of employees are increased, which enables a more efficient activity in similar projects in the future. On the other hand, competencies that are not used decrease

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over time, both as a consequence of a reduction of efficiency by lack of routine and by technological progress that makes parts of the acquired knowledge obsolete; the latter effect is usually captured by the term “knowledge depreciation”. Management paradigms such as the *resource-based view of the firm* (Wernerfelt 1984; Peteraf 2001) implicitly address questions of competence development, but models for representing this effect in quantitative terms and for optimizing decisions in view of it are still rather rare. Since especially in the area of R & D, and also in other branches of project management, a consideration of growth (or decrease) of competencies is crucial for appropriate decision making, it would be desirable to shed more light on the dynamic interplay between project selection decisions and competence development by a theoretical model that allows to *understand* the key features of the process. In addition, the popular recommendation to concentrate on “core competencies” seems still to be insufficiently justified on the level of theoretical analysis; it is not clear that to which extent the claimed superiority of such a policy can be explained within a rational decision making paradigm, and what the limits of an exclusive focus on core competencies are.

In contrary to works on project selection considering competence *development*, there is a rather large number of articles taking into account different skills of the employees in a *static* view. Some examples are Bassett (2000) who assigns projects or tasks to employees in an R & D application in the chemical industry based on their skills, Bellenguez and Neron (2005) who extend the classical resource-constrained project scheduling problem (RCPSp) to a multi-skill project scheduling problem, Techawiboonwong et al. (2006) who determine schedules for assigning temporary industry workers to flow lines, Alba and Chicano (2007) who put up an RCPSp formulation for applications in software engineering, Eiselt and Marianov (2008) who represent employees and tasks in a skill space and assign employees to tasks in such a way that employee–task distances are minimized, or Yoshimura et al. (2006) who treat the combined problem of selecting projects and allocating human resources to them based on employee’s skills and motivation.

Models for *learning* effects related to competencies have been investigated, e.g., by Wu and Sun (2006), Ngwenyama et al. (2007), Pendharkar and Subramanian (2007), Armbruster et al. (2007), Fioretti (2007), or Heimerl and Kolisch (2008a). A rather small group of articles not only considers learning, but also *forgetting* or knowledge depreciation. For example, Argote and Epple (1990) and Darr et al. (1995) present a model for acquisition and depreciation of knowledge as well as for knowledge transfer, building on the traditional concept of the learning curve. Chen and Edgington (2005) provide a model for knowledge creation in firms describing the dependence between the time invested into a process and the achieved competence by a logistic function, and they also take account of knowledge depreciation by introducing a depreciation rate. Süer and Tummaluri (2008) use a model for learning and forgetting of skills in order to improve assignments of operators to operations in cellular manufacturing. In Gutjahr et al. (2007, 2008a,b) and Stummer et al. (2009), an integrated model combining project selection, staff assignment, and time scheduling with aspects of competence development is introduced, where competence is seen both as a prerequisite for immediate economic efficiency of production, and as a strategic objective relating to the future market position of the firm. Heimerl and Kolisch (2008b) use a model for learning and forgetting similar to that in Gutjahr et al. (2008a), but do not deal

with the aspects of project selection and time scheduling; on the other hand, they also investigate the possibility of outsourcing certain tasks, which requires a consideration of the tradeoff between internal and external costs. Moreover, they address the question whether competence development goals, formulated in their model as constraints, will lead to specialization of individual employees and/or of the entire firm.

The indicated articles providing models for learning and forgetting do not consider the aspect of project portfolio selection, with the exception of Gutjahr et al. (2007, 2008a,b) and Stummer et al. (2009), where binary decisions on either to take or to reject a candidate project are assumed. This is a frequent view in the project portfolio selection literature, but there is also a possible alternative, as represented, e.g., by Loch and Kavadias (2002), where dynamic project portfolio selection is studied on the assumption of *continuous* amounts of investments into projects rather than of all-or-nothing decisions.

In the present paper, we adopt the continuous model for project selection. Our main goal is to derive theoretical results on the process of optimal project portfolio selection over time, taking explicit account of competence development by learning and forgetting. After setting up suitable models for this process, general mathematical properties will be derived. Furthermore, also the question of how to compute optimal policies numerically will be addressed.

In particular, it will be shown that when compared to financial portfolio management which uses the advantages of *diversification* in a way that can be analyzed by the tools of classical portfolio theory, *project* portfolio management is characterized by an (additional) force aiming at *concentration*. The two opposite forces lead to a typical tradeoff which has been addressed in the empirical literature by terms such as the “innovator’s dilemma” (Christensen 2003).

The outline of the paper is the following: in Sect. 2, a basic model, where competencies are directly related to project classes, will be presented and discussed. Section 3 provides analytical results concerning optimal policies. The numerical computation of optimal policies is treated in Sect. 4. In Sect. 5, the model is generalized to the situation where a project class may require different competencies to possibly differing degrees. Section 6, finally, contains concluding remarks and topics for further research.

2 The basic model

We start with the description of a special, still relatively simple model in Sect. 2 and will generalize it to a more flexible problem formulation later. A set of *project classes* indexed by $i = 1, \dots, n$, is assumed as given. It is supposed that each project class i provides the opportunity for investments into projects of basically the same type to an arbitrary extent that can also vary over time. Time is discretized by decomposition into *periods* $t = 1, \dots, T$. The planning horizon T is considered as fixed. We assume that the degree of investment into project class i is a *continuous* rather than a discrete variable, which means that projects can also be partially funded. We do not distinguish between different projects assigned to the same project class i ; only the overall amounts of engagement in project classes 1, 2, etc. are relevant in the considered model.

Furthermore, we suppose that a fixed team of employees is available for the execution of the selected projects. By the term *competencies*, we refer to abilities, skills etc. of

the employees. Competencies improve by learning, experience, and “training-on-the-job”, but can also decrease by forgetting or knowledge depreciation, if no activities requiring them are performed over some time. For shortness, we shall use only the terms *learning* and *forgetting* when referring to these two antagonistic phenomena in the sequel.

In this paper, we always treat competencies as aggregated over the entire team, i.e., a competency is supposed to have a certain value for the *firm*. (A refined consideration, as it has been proposed in Gutjahr et al. (2008a,b), would take the distribution of competencies among the individual employees into account.) In the basic model presented in this section, we assume that each project class i requires a specific (core) competency, which can then be denoted by i again. This assumption, which is also implicit in Chen and Edgington (2005), will be relaxed in the generalized model of Sect. 5, where we assume that a list of competencies, indexed by $r = 1, \dots, R$, is given, and that each project class i can require one or more competencies from this list, each according to a certain, pre-defined percentage.

Competencies are quantified by *competence scores* z_{it} , where i is the index of the considered competency, t refers to the time period, and z_{it} is a real (possibly negative) value on some fixed scale. We suppose that the competence score z_{it} is either measured by well-known methods of organizational psychology, or estimated based on expert judgements. Work time of amount x in project class i (requiring competency i) during a certain period is assumed to increase the current competence score by an increment proportional to x (“learning”). On the other hand, forgetting is assumed to reduce the competency score by a fixed decrement per period; this reduction can of course be (over-)compensated by the learning effect mentioned before.

From a specific competence score of the team, we derive the team’s *efficiency* value in the respective competency by applying a monotonous transformation of a similar type as the so-called *learning curves* described in literature; often, the *logistic function* has been used for modeling dependencies of this kind (Chen and Edgington 2005; Ngwenyama et al. 2007). The efficiency value γ_{it} of the team in a certain competency i is a real value between 0 and 1 defined as the share of effective work the considered team can perform during one time unit in projects requiring competency i , compared to the effective work performed during one time unit by a team with “perfect” competency in i . In other words, if a team with perfect competency in i needs d time units to complete a project from project class i , then a team with efficiency $\gamma_{it} \in [0, 1]$ needs d/γ_{it} time units to complete the same project. We distinguish between the “effective work time” required by this project, which is d for both teams, and the “real work time”, which is d/γ_{it} for the team with efficiency γ_{it} , and d for the team with efficiency 1. The relation between competence score z and efficiency γ is given by the non-decreasing function φ mapping the set of reals R into the interval $[0, 1]$: for each competency i and each period t , we assume $\gamma_{it} = \varphi(z_{it})$. As stated above, a suitable function type for φ would be that of the logistic (or sigmoid) function given by

$$\varphi(z) = [1 + a \exp(-bz)]^{-1}, \quad (1)$$

but also other non-decreasing functions can be used. We call φ the *learning function*.

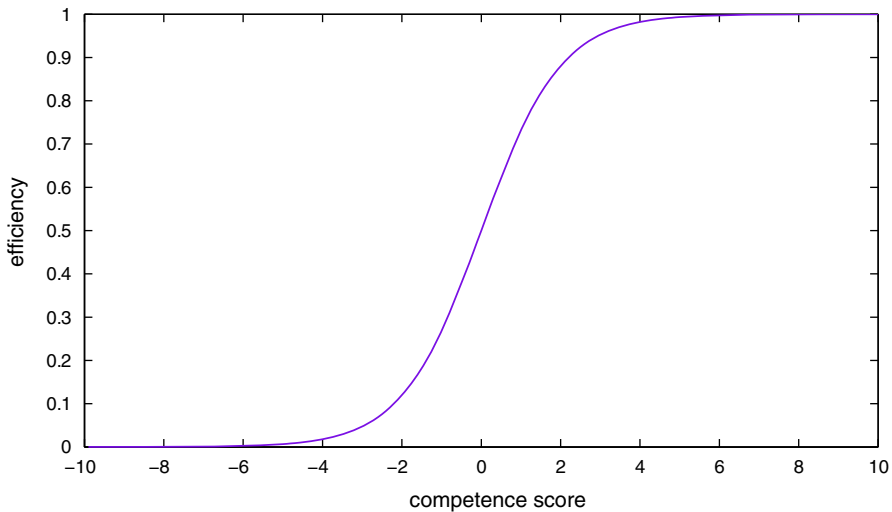


Fig. 1 An example for a possible shape of the learning function φ mapping competence scores to efficiency values. Here, the logistic function with parameters $a = b = 1$ is used

An example is shown in Fig. 1 (logistic function with $a = b = 1$). In this example, the competence scores are scaled in such a way that positive scores correspond to efficiency values larger than 50%, whereas negative scores correspond to efficiency values lower than 50%. As it is seen, the competence score can fall to arbitrarily low values; the corresponding efficiency, however, will never fall below zero, although it can come arbitrarily close to zero. Similarly on the positive axis: by learning, the competence score can be increased to an arbitrary degree, but the marginal gains in efficiency will diminish, such that an efficiency value of unity can never be exceeded.

The logistic learning function has a convex part (for lower competence scores) and a concave part (for higher competence scores). Although the dependence between competence score and efficiency does not necessarily have this type of shape, one would expect that an increment of experience (expressed by an increment of the competence score) will have the largest impact on the growth of efficiency if the current competence score is in some medium range. If the current score is very low, the person is not able to exploit new experience in the best-possible way, because important basic skills are missing. On the other hand, if the current score is very high, new experience will not increase the efficiency anymore to a large extent, since it has already almost reached its upper limit. Thus, S-shaped learning functions seem to be most realistic.

By x_{it} , we denote the share of the overall work capacity of the team that is invested into project class i during period t . The real-valued variables x_{it} are the decision variables in our model. It is convenient to choose the work time unit as the overall amount of work time the team can invest during one period. Then, x_{it} can alternatively be interpreted as the amount of work time invested into project class i during period t .

What is still required to fully specify the model is information on economic gains (profits). We assume that each unit of *effective* work time invested into projects from project class i yields a profit of $w_i > 0$ currency units. For the sake of simplicity,

in the present version of our model, we do not discount the achieved profits in the single periods when cumulating them to the overall profit. (An extended model could discount them according to a certain rate by period, leading to a net present value.) The values w_i represent the current market situation which we assume as constant over the planning horizon. The situation where the market situation might change will be discussed later.

In a mathematical representation, our optimization model can be formulated as follows:

$$\sum_{i=1}^n w_i y_i \rightarrow \max \quad (2)$$

$$\gamma_{it} = \varphi(z_{it}) \quad \forall i, t \quad (3)$$

$$z_{it} = z_{i1} - \beta_i(t-1) + \eta_i \sum_{s=1}^{t-1} x_{is} \quad \forall i, t \quad (4)$$

$$\sum_{i=1}^n x_{it} = 1 \quad \forall t \quad (5)$$

$$\sum_{t=1}^T \gamma_{it} x_{it} = y_i \quad \forall i \quad (6)$$

$$x_{it} \geq 0 \quad \forall i, t \quad (7)$$

$$y_i \geq 0 \quad \forall i. \quad (8)$$

Equation (2) represents the objective function as the sum of the profits $w_i y_i$ from project classes i over all i ; therein, y_i denotes the overall effective work time in project class i . Equation (3) specifies the dependence between competence scores and efficiencies. By Eq. (4), the increment or decrement of the competence scores due to learning or forgetting is expressed: the parameter β_i gives the *depreciation constant*, i.e., the amount by which the score of competency i decreases during one period, if this effect is not (over-)compensated by exercising the respective competence. The parameter η_i , on the other hand, gives the *learning constant*, i.e., that amount by which the score of competency i increases due to work during a unit of (real) work time. Note that $\sum_{s=1}^{t-1} x_{is}$ is the cumulated real work time in competency i during the periods preceding period t . Since the maximum amount of work that can be performed in a competency i during one period is unity, $\beta_i \leq \eta_i$ must hold for each i ; otherwise, competency i would necessarily diminish over time in spite of learning. A value of β_i given by, say, $\beta_i = 0.1 \eta_i$, just means that if a competency has been increased by a certain amount within one month of exclusive work within this competency, then it takes 10 months until the score in this competency has fallen back again to the original level by forgetting or knowledge depreciation. Equation (5) ensures that the sum over all work time investment during a period equals the overall capacity of the team during one period, which is unity by our choice of the work time scale. Via Eq. (6), we express y_i , the overall effective work time invested into project class i during the planning horizon, as the sum of the effective work times (efficiencies multiplied by

real work times) over all periods. Equations (7) and (8) give nonnegativity constraints for real work times and for investments, respectively.

By inserting (3) together with (4) into (6) and using the resulting expression for re-writing (2), we obtain the more concise formulation

$$\sum_{i=1}^n w_i \sum_{t=1}^T x_{it} \varphi \left(z_{i1} - \beta_i(t-1) + \eta_i \sum_{s=1}^{t-1} x_{is} \right) \rightarrow \max \quad (9)$$

subject to

$$\mathbf{x}_t \in S_n \quad (t = 1, \dots, T). \quad (10)$$

Therein, $\mathbf{x}_t$ denotes the column vector $(x_{1t}, \dots, x_{nt})^T$, and $S_n = \{\mathbf{x} \in \mathbb{R}^n \mid \sum_{i=1}^n x_i = 1\}$ is the simplex in the space $\mathbb{R}^n$.

There are close relations of the model above to that in [Gutjahr et al. \(2008a\)](#). The main differences are:

1. Instead of a 0–1-decision whether or not a project should be funded, we allow partial or even “surplus” funding.
2. We consider a homogeneous team, i.e., we do not take account of individual competencies.
3. In the model of the present paper, competency gains are not taken into account via a specific term in the objective function, but rather indirectly by the fact that they increase the efficiency and thus enable a more profitable work in the future.
4. Team capacities are assumed as identical over all periods, and there are no ready times nor due dates.
5. At least in the simpler version of the model treated in this section, each project requires exactly one specific competency. (This assumption will be relaxed later.)

Differences 2–5 are restrictions to special cases. This reduction of the complexity will allow a deeper mathematical insight into the problem structure, compared to [Gutjahr et al. \(2008a\)](#). Difference 1 is a fundamental modification effecting that a continuous instead of a discrete optimization problem is considered. Whether the discrete or the continuous situation is more appropriate depends on the specific application.

In principle, the model can be solved by Dynamic Programming (DP), although the continuous structure of the solution vectors $\mathbf{x}_t$ poses some problems for DP. Borrowing from common terminology in DP, we will call a solution $(\mathbf{x}_1, \dots, \mathbf{x}_T)$ a *policy*.

3 Analytical results

Throughout the paper, we use the following notation: $\mathbf{e}_i$ denotes the i th extremal point of the simplex S_n , which is given by the i th unit vector $(0, \dots, 0, 1, 0, \dots, 0)^T$. Therein, the j th component of $\mathbf{e}_i$ is δ_{ij} with $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ otherwise.

Let us start with a short outline of the case where the investment vector $\mathbf{x}_t \in S_n$ is kept unchanged throughout the entire planning horizon, i.e., $\mathbf{x}_t = \mathbf{x}$ for all $t = 1, \dots, T$. We show that in this case, it is optimal to invest all available capacities

into one single project class. The result only needs the very mild assumption that the learning function φ mapping competence scores to efficiencies is nondecreasing. The proofs of all theorems can be found in Appendix A.

Theorem 1 *For a nondecreasing learning function φ , the function*

$$G(\mathbf{x}) = \sum_{i=1}^n w_i x_i \sum_{t=1}^T \varphi(z_{i1} + (\eta_i x_i - \beta_i)(t-1)) \quad (11)$$

obtained from (9) by setting $\mathbf{x}_t = \mathbf{x} = (x_1, \dots, x_n)^T$ for all t , takes a maximum in one of the extremal points $\mathbf{e}_i$ ($i = 1, \dots, n$) of the simplex S_n . In informal terms: if investments have to be kept constant over time, it is optimal to put all efforts into one of the project classes and none into the other project classes.¹

Now, we turn to the more interesting case where policies that vary the investment vector $\mathbf{x}_t \in S_n$ from period to period are admitted. It will be shown that in this scenario, there are still special conditions under which it is always optimal to put, in each single period, all efforts into only one project class. To fix notation, let us write X for the matrix of the elements x_{it} ($i = 1, \dots, n$; $t = 1, \dots, T$), i.e., $X = (\mathbf{x}_1, \dots, \mathbf{x}_T)$. Moreover, let

$$G_t(X) = \sum_{i=1}^n w_i x_{it} \varphi \left(z_{i1} - \beta_i(t-1) + \eta_i \sum_{s=1}^{t-1} x_{is} \right), \quad (12)$$

such that the objective function

$$G(X) = \sum_{i=1}^n w_i \sum_{t=1}^T x_{it} \varphi \left(z_{i1} - \beta_i(t-1) + \eta_i \sum_{s=1}^{t-1} x_{is} \right) \quad (13)$$

from (9) can be expressed as

$$G(X) = \sum_{t=1}^T G_t(X). \quad (14)$$

Note that $G_t(X)$ actually only depends on the column vectors $\mathbf{x}_s$ in X with $s \leq t$. For this reason, we will also write $G_t(\mathbf{x}_1, \dots, \mathbf{x}_t)$ for $G_t(X)$. For fixed $\mathbf{x}_1, \dots, \mathbf{x}_{t-1}$, the function $G_t(X)$ is linear in $\mathbf{x}_t$.

Theorem 2 *Suppose that the values z_{i1} , β_i and η_i are such that during all periods, the arguments of the learning function φ remain in a subinterval $[c_1, c_2] \subseteq \mathbb{R}$ in which*

¹ Typically, “extremal” solutions occur as maxima in cases where the objective function is convex. It should be observed that for the result above, we do not need a convexity condition on $\varphi(z)$ nor on the entire function $G(\mathbf{x})$.

φ is convex. Then there is an optimal policy $(\mathbf{x}_t)$ ($t = 1, \dots, T$) choosing in each period t an extremal point of the simplex S_n as the current investment vector.²

Example 1 Let us consider the logistic learning function φ with parameters $a = b = 1$ depicted in Fig. 1. Suppose that the considered firm is relatively new in business and has therefore competence scores below -1 in all competencies (in the chosen scaling). Furthermore, let us suppose that $\eta_i \leq 1/T$ for all i , such that all competence scores will remain non-positive until the end of the planning horizon. Then the assumptions of Theorem 2 are satisfied, and the theorem claims that in each period, all efforts should be focused on a single project class (which may depend on the period). A policy with a mixed project portfolio in some period cannot be superior.

Remark 1 Theorem 2 reminds of recommendations of strategies where the firm always fully invests into its *core competency* [cf. Hamel and Prahalad (1994) for other approaches concerning the focus that companies should put into competencies, see, e.g., Hitt and Ireland (1985), Henderson and Cockburn (1994) and Henderson (2004)]. From the viewpoint of our theoretical result, however, three remarks have to be added:

- The competency leading to the maximum gain is not yet completely defined by the competencies available at the beginning of the planning interval. Rather than that, the relations of these initial competencies to the achievable profits in the corresponding project classes have to be taken into account. In other words, the view cannot be exclusively “resource-based” (cf. Wernerfelt 1984), but has to address the aspect of the *demand* as well.
- In the case of a logistic (or, more generally, S-shaped) learning function, the optimality of investing all available human resources into a single project class can only be confirmed for the outlined *first* phase in the development of a company where the marginal returns in terms of efficiency are still increasing with growing engagement in projects of a certain project class. It is not clear what will happen in later phases. If there are competencies for which the firm is in a “distinctly concave” part of the learning function (e.g., the range around the scores 1 to 3 in Fig. 1), diversification may pay off.
- Our model is deterministic. From general portfolio theory, it is well-known that another good reason for diversification is a situation where essential parameters are unknown, such that a decision under risk has to be made (cf. the discussion in the Sect. 6).

Remark 2 One might conjecture that if the learning function φ is concave instead of convex, problem (9)–(10) is a concave maximization problem and has therefore a unique local maximizer on the simplex. Simple counter-examples (which we omit here), however, show that this is not true: concavity of φ does neither imply that objective function (9) is concave, nor that the optimal solution must be unique.

Contrary to the scenario of fixed investments over time (see the previous section), in the scenario of variable investments, the condition of Theorem 2 concerning the

² In general, the extremal points $\mathbf{x}_t$ chosen in the T periods need not to be identical. Under special conditions, a change of the optimal project class can be excluded (see Theorem 3 below).

convexity of the learning function in the range of interest is essential indeed. The following example shows that if this condition is omitted, optimal policies need not to be extremal anymore:

Example 2 Let φ be given as the following non-decreasing, piecewise linear function:

$$\varphi(z) = \begin{cases} 0, & \text{if } z < -1, \\ (1+z)/2 & \text{if } -1 \leq z \leq 1, \\ 1 & \text{if } z > 1. \end{cases} \quad (15)$$

This function is concave on the set of positive reals. Consider $n = 2$ project classes and $T = 2$ periods, and assume initial competence scores of $z_{i1} = 0$ for $i = 1, 2$. Moreover, suppose that $\beta_1 = \beta_2 = 0$ (i.e., no forgetting takes place), $\eta_1 = 2$, and $\eta_2 = 0$. Thus, competency 1 can easily be increased by learning, whereas competency 2 cannot be increased. Concerning the profits, we assume $w_1 = 1$ and $w_2 = 3/2$. Thus, project class 2 is somewhat more profitable than project class 1.

We compute the optimal policy X by means of the auxiliary functions G_t , applying (12)–(14). In our case of $n = 2$ projects, the vector $\mathbf{x}_t$ is already described by x_{1t} , since $x_{2t} = 1 - x_{1t}$. Equations (12)–(14) yield $G(X) = G_1(X) + G_2(X) = G_1(\mathbf{x}_1) + G_2(\mathbf{x}_1, \mathbf{x}_2)$ with

$$\begin{aligned} G_1(\mathbf{x}_1) &= w_1 x_{11} \varphi(z_{11}) + w_2 x_{21} \varphi(z_{21}) \\ &= w_1 x_{11} \varphi(z_{11}) + w_2 (1 - x_{11}) \varphi(z_{21}) = 3/4 - (1/4)x_{11} \end{aligned} \quad (16)$$

and

$$\begin{aligned} G_2(\mathbf{x}_1, \mathbf{x}_2) &= w_1 x_{12} \varphi(\eta_1 x_{11}) + w_2 x_{22} \varphi(\eta_2 x_{21}) \\ &= w_1 x_{12} \varphi(2x_{11}) + w_2 (1 - x_{12}) \varphi(0) = x_{12} \varphi(2x_{11}) + (3/4)(1 - x_{12}). \end{aligned} \quad (17)$$

We start with considering $\mathbf{x}_1$ as fixed and optimize the contribution $G_2(\mathbf{x}_1, \mathbf{x}_2)$ with respect to $\mathbf{x}_2$ for this fixed $\mathbf{x}_1$. If $\mathbf{x}_1$ is viewed as constant, (17) becomes a linear function of $\mathbf{x}_2 = (x_{12}, 1 - x_{12})$ which takes its maximum in one of the extremal points $\mathbf{e}_1$ or $\mathbf{e}_2$. From $G_2(\mathbf{x}_1, \mathbf{e}_1) = \varphi(2x_{11})$, $G_2(\mathbf{x}_1, \mathbf{e}_2) = 3/4$ and (15) we immediately derive

$$\max_{i=1,2} G_2(\mathbf{x}_1, \mathbf{e}_i) = \begin{cases} \varphi(2x_{11}), & \text{if } x_{11} \geq 1/4, \\ 3/4, & \text{if } x_{11} < 1/4. \end{cases}$$

Now let us make $\mathbf{x}_1$ variable. We have then to maximize

$$F(\mathbf{x}_1) = G_1(\mathbf{x}_1) + \max_{i=1,2} G_2(\mathbf{x}_1, \mathbf{e}_i).$$

In the case of $x_{11} < 1/4$, we find $F(\mathbf{x}_1) \leq 3/2$, whereas in the case of $x_{11} = 1/2$, a higher value of $F(\mathbf{x}_1)$ can be achieved, namely $13/8$. Thus, $3/4 - (1/4)x_{11} + \varphi(2x_{11})$ has to be maximized for $x_{11} \geq 1/4$. For x_{11} exceeding the value $1/2$, the term $\varphi(2x_{11})$

does not grow anymore, while $3/4 - (1/4)x_{11}$ decreases. Hence, we can restrict ourselves to the interval $x_{11} \in [1/4, 1/2]$, where $\varphi(2x_{11}) = 1/2 + x_{11}$. One immediately sees that the optimal solution is $x_{11}^* = 1/2$. Together with $x_{12}^* = 1$, this gives $(x_{11}^*, x_{12}^*) = (0.5, 1)$ as the optimal policy. It is easy to verify that each *extremal* policy, e.g., $(0, 1)$ or $(1, 1)$, performs worse than the policy $(0.5, 1)$ using a mixed portfolio in the first period.

It might be conjectured that at least in cases where the optimal policy consists only of extremal points of S_n (i.e., where each $\mathbf{x}_t$ is one of the unit vectors of $\mathbb{R}^n$), all these points must be identical, such that in each period, a decision is made for the same project class. We can show that this holds indeed in the specific situation where no *forgetting* takes place. If, in this situation, the problem is further restricted by the additional constraint that all x_{it} must be integers (which is equivalent to admitting only extremal points of S_n as partial solutions for the single periods), then an optimal policy does not change the project class into which efforts are invested:

Theorem 3 Assume that $\beta_i = 0$ for all i (no forgetting), that $\eta_i > 0$ and $w_i > 0$ for all i , and that the learning function φ is strictly increasing. Then an optimal solution of the problem (9)–(10) with the additional constraint that all x_{it} are integers (investment into only one project class in each period) is of the form $(e_i^*, \dots, e_i^*)$, i.e., it consists of identical investment decisions for each period.

We may interpret Theorem 3 by saying that in a scenario where knowledge does not depreciate and employees always preserve their competencies once acquired, a manager who has already decided to follow a “focused” strategy of project portfolio selection in each single period can restrict herself/himself to a *static* decision: s/he has only to find out what the most profitable project class is and can stick to it then as long as external conditions do not change.³ Nevertheless, as Example 2 shows, s/he may be worse off in this way than a manager allowing mixed portfolios.

Moreover, the assumption that $\beta_i = 0$ for all i is essential for the validity of Theorem 3. As the following example shows, in the case where forgetting takes place, also under the additional constraint that all x_{it} are integers (investment into only one project class per period possible), the optimal decisions in the single periods need not to be the same.

Example 3 Consider again the function φ defined by (15). Let $n = 2$, $T = 2$, $z_{11} = 1$, $z_{21} = 0$, $\beta_1 = 3$, $\beta_2 = 0$, $\eta_1 = 4$, $\eta_2 = 1$, and $w_1 = w_2 = 1$. Now

$$G_1(\mathbf{x}_1) = w_1 x_{11} \varphi(z_{11}) + w_2 x_{21} \varphi(z_{21}) = x_{11} + (1 - x_{11}) \cdot (1/2) = 1/2 + (1/2) \cdot x_{11},$$

³ One might argue that in a strict sense, the case $\beta_i = 0$ without any knowledge depreciation does not occur in practice. However, this case is nevertheless of considerable interest as the boundary case approximating the situation where the depreciation constants β_i are small compared to the learning constants η_i . Such a situation is frequent since typically, knowledge depreciation happens on a slower time scale than knowledge accumulation. It seems to be for this reason that the majority of publications on learning in project portfolio management does not consider forgetting or knowledge depreciation in their models at all. The most realistic assumption possibly consists in viewing some competencies, related to knowledge that depreciates rather in periods of years than of months, as having low β_i values that are often negligible compared to the η_i values, whereas other competencies, related to fast-evolving areas, have β_i values that cannot be neglected.

and

$$\begin{aligned} G_2(\mathbf{x}_1, \mathbf{x}_2) &= w_1 x_{12} \varphi(z_{11} - \beta_1 + \eta_1 x_{11}) + w_2 x_{22} \varphi(z_{21} - \beta_2 + \eta_2 x_{21}) \\ &= x_{12} \varphi(-2 + 4x_{11}) + (1 - x_{12}) \varphi(x_{21}) = (1 - x_{12}) \varphi(x_{21}). \end{aligned}$$

Obviously, for maximizing $G_2(\mathbf{x}_1, \mathbf{x}_2)$, it is always better to choose $x_{12} = 0$, independently of x_{21} . For maximizing $G_1(\mathbf{x}_1)$, the best choice is $x_{11} = 1$. Therefore, the optimal policy is $(\mathbf{e}_1, \mathbf{e}_2)$, and both $(\mathbf{e}_1, \mathbf{e}_1)$ and $(\mathbf{e}_2, \mathbf{e}_2)$ are strictly worse.

Example 4 Let us study special cases of functions φ that are either convex or concave in the range of interest in more detail. Perhaps the simplest class of such functions is the following: As in Theorem 2, suppose that the values z_{i1} , β_i and η_i are such that during all periods, the arguments of the learning function φ remain in a subinterval $[c_1, c_2] \subseteq \mathbb{R}$. Now, for some $\alpha \in \mathbb{R}_0^+$, define

$$\varphi(z) = \left(\frac{z - c_1}{c_2 - c_1} \right)^\alpha.$$

For $\alpha > 1$, $\alpha = 1$ and $\alpha < 1$, this function is convex, linear and concave, respectively. Since the interval $[c_1, c_2]$ can be mapped to the interval $[0, 1]$ by a simple linear transform, we can confine ourselves to the special case $c_1 = 0$ and $c_2 = 1$ without loss of generality, setting $\varphi(z) = z^\alpha$ and assuming that during all periods, competence scores remain values in $[0, 1]$. In this special case, Eqs. (9)–(10) can be solved by using the Karush–Kuhn–Tucker conditions. Exploiting the fact that in the last period T , an extremal decision is optimal, can speed up the computation. In the following, we derive explicit formulas for the solution in the special case $n = 2$ and $T = 2$.

As in Example 2, we maximize

$$F(\mathbf{x}_1) = G_1(\mathbf{x}_1) + \max_{i=1,2} G_2(\mathbf{x}_1, \mathbf{e}_i) = \max(\psi_1(\mathbf{x}_1), \psi_2(\mathbf{x}_1))$$

with $\psi_i(\mathbf{x}_1) = G_1(\mathbf{x}_1) + G_2(\mathbf{x}_1, \mathbf{e}_i)$ ($i = 1, 2$) for $\mathbf{x}_1 = (x_{11}, 1 - x_{11}) \in S_2$, i.e., $x_{11} \in [0, 1]$. The optimal value x_{11}^* of x_{11} is either taken at the boundary of the interval $[0, 1]$, i.e., in one of the points 0 or 1, or in $]0, 1[$. In the last case, it follows from the continuity of the functions ψ_i that x_{11}^* must either be a local maximum of ψ_1 or of ψ_2 (or of both). Thus, to solve the problem, it suffices to compute the local maxima $x_{11}^{(1)}$ and $x_{11}^{(2)}$ of ψ_1 and ψ_2 , respectively. They are only of relevance if they happen to lie within the interval $]0, 1[$. Those of them that satisfy the last condition are candidates for x_{11}^* , as well as the points 0 and 1. All that remains to do is to take that candidate that yields the best value of $F(\mathbf{x}_1)$. This gives the optimal decision for period 1. Whether ψ_1 or ψ_2 yields a larger value at x_{11}^* determines the optimal decision for period 2 ($\mathbf{e}_1$ or $\mathbf{e}_2$, respectively).

For determining $x_{11}^{(i)}$, we have to find the value of x_{11} where the first derivative $\frac{d}{dx_{11}} \psi_i(x_{11}, 1 - x_{11})$ vanishes, and to check the second derivative at this value.

Straightforward computation yields the following roots of the first derivatives:

$$x_{11}^{(1)} = \frac{1}{\eta_1} \left\{ \left[\frac{w_2 z_{21}^\alpha - w_1 z_{11}^\alpha}{w_1 \eta_1 \alpha} \right]^{1/(\alpha-1)} - z_{11} + \beta_1 \right\} \quad (18)$$

and

$$x_{11}^{(2)} = 1 - \frac{1}{\eta_2} \left\{ \left[\frac{w_1 z_{11}^\alpha - w_2 z_{21}^\alpha}{w_2 \eta_2 \alpha} \right]^{1/(\alpha-1)} - z_{21} + \beta_2 \right\}. \quad (19)$$

Moreover, we find

$$\frac{d^2}{dx_{11}^2} \psi_i(x_{11}, 1 - x_{11}) = \begin{cases} w_1 \alpha (1 - \alpha) \eta_1^2 (z_{11} - \beta_1 + \eta_1 x_{11})^{\alpha-2}, & i = 1, \\ w_1 \alpha (1 - \alpha) \eta_2^2 (z_{21} - \beta_2 + \eta_2 (1 - x_{11}))^{\alpha-2}, & i = 2. \end{cases}$$

The sign of the second derivative depends on the special value of α . Note that for each $z > 0$, the term $z^{\alpha-2} = \exp((\alpha - 2) \log z)$ is strictly positive.

- (a) In the case $\alpha > 1$ (convex case), $\alpha(1 - \alpha) > 0$. Hence, the second derivative is nonnegative on $]0, 1[$ in this case, which means that an “inner” local maximum $x_{11} \in]0, 1[$ cannot exist. This is in full agreement with Theorem 2.
- (b) In the case $\alpha < 1$ (concave case), $\alpha(1 - \alpha) < 0$. Hence, provided that $\eta_i > 0$, the second derivative of $\psi_i(x_{11}, 1 - x_{11})$ is negative on $]0, 1[$ in this case, with the consequence that each root (18) or (19) corresponds to a local maximum.

Summarizing, we can state that the possible local maxima are the points $0, 1, x_{11}^{(1)}$ (if in $]0, 1[$) and $x_{11}^{(2)}$ (if in $]0, 1[$).

As a concrete special case, we may take an example with a similar flavor as Example 2: set $w_1 = 1$, $w_2 = 1.1$, $z_{11} = 0.03$, $z_{21} = 0.02$, $\beta_1 = \beta_2 = 0.01$, $\eta_1 = 0.90$, $\eta_2 = 0.02$, and $\alpha = 0.05$ (concave learning function). Comparing the four solution candidates indicated above, we find that the optimal solution for the first period is $x_{11}^* = x_{11}^{(1)} = 0.7275$. Since this solution is achieved via ψ_1 , the optimal solution for the second period is e_1 . Let us give an interpretation of this outcome: In period 1, it is advantageous to use part of the capacity for increasing the leading competency 1, which is simultaneously much easier to develop. However, because of the diminishing marginal returns in the concave case, this competency should not be extended up to its maximal achievable degree by fully investing into project class 1. Rather than that, only a certain share of the capacity should be used for competence development; the remaining part should be spent for maximizing the *immediate* gain in period 1. Since project class 2 is more profitable, this remainder is to be invested into project class 2.

In the convex case $\alpha > 1$, we have increasing instead of diminishing marginal returns from competence development, which excludes situations of the above type where a portfolio split provides advantages.

Remark It may be interesting to compare the model for competence learning and forgetting applied in the present paper with previous models from the literature, e.g., with

the model presented in Argote and Epple (1990) and Darr et al. (1995). This model builds on the conventional learning curve

$$y = ax^{-b}, \quad (20)$$

where y , a , x and b denote the effort for producing the x th unit, the effort for producing the first unit, the cumulative number of units produced, and a parameter expressing effort reductions by learning, respectively. To take account of knowledge depreciation, Argote, Epple and Darr use a special discounting model assuming that in each period, only part of the acquired knowledge persists, whereas the other part gets lost by depreciation. In Appendix B, we argue that the *learning* component of this model corresponds in some sense to the special case of the power learning function $\varphi(z) = z^\alpha$ studied in Example 4 above. The knowledge depreciation component would be more difficult to capture in terms of our model, since it applies a different mechanism than the learning component, whereas in our model, these two aspects are treated in a conceptually related way.

Let us now drop the integrality constraint and return to the more general context where it is allowed to distribute the effort in every period over several project classes. In the specific case of Example 2, nevertheless, it turned out that focusing on a single project was optimal at least in the *last* period. This observation can be generalized: a portfolio split in the last period is never advantageous. We can even show a more general result asserting that essentially, the number of project classes on which the investment should be focused reduces by one from period to period, ending with an extremal decision in the last period, where only a single project should be considered. To formulate this result in precise terms, we introduce the following definition: By the *support* of some $\mathbf{x}_t$, we understand the set of indices of project classes to which $\mathbf{x}_t$ assigns nonzero efforts. Formally:

$$\text{support}(\mathbf{x}_t) = \{i \in \{1, \dots, n\} : x_{it} > 0\}. \quad (21)$$

Moreover, for $X = (\mathbf{x}_1, \dots, \mathbf{x}_T)$ and each $t = 1, \dots, T$, we consider the set

$$\text{support}_t^+(X) = \bigcup_{\tau=t}^T \text{support}(\mathbf{x}_\tau)$$

of indices of project classes into which X invests nonzero efforts in period t or later. Evidently,

$$\text{support}_1^+(X) \supseteq \text{support}_2^+(X) \supseteq \dots \supseteq \text{support}_T^+(X).$$

For a set A , the number of elements in A is denoted by $|A|$. With this notation, the following general property can be shown:

Theorem 4 Problem (9)–(10) has an optimal solution $X^* = (\mathbf{x}_1^*, \dots, \mathbf{x}_T^*)$ with

$$|\text{support}_t^+(X^*) \setminus \text{support}_{t+1}^+(X^*)| \leq 1 \quad (t = 1, \dots, T-1) \quad (22)$$

and

$$1 \leq |\text{support}_t^+(X^*)| \leq T - t + 1 \quad (1 \leq t \leq T). \quad (23)$$

In less formal terms: x_T^* only invests into a single project class; x_{T-1}^* can invest into this class and in at most one additional class; x_{T-2}^* can invest into these two classes and in at most one additional class, etc.

Theorem 4 indicates that if a firm starts with a broad portfolio of project classes, it should successively restrict the set of active project classes and continue this process of specialization until only a single project class remains active. In general, if an optimal path is followed, reaching complete specialization on one single project class can take more than one period (cf. Example 2).

For a realistic interpretation of the result, however, it is important to note that in practice, two additional aspects that are not represented in our model can possibly counteract the mentioned tendency of an optimal policy to increasing specialization: First, uncertainty on profits may make diversification more attractive to a risk-averse decision maker (cf. Remark 3 after Theorem 2). Second, also a risk-neutral decision maker may be faced with changes in her/his estimates of future profits: Already at the beginning of period 2, a re-estimation of the profits w_i assigned to project classes $i = 1, \dots, n$ may yield changed values $\bar{w}_i$, and under these values, the original policy must also be adapted. This can force an increase in diversification. For example, even if it would have been optimal to focus on one and the same project class i_0 throughout periods $t = 1, \dots, T$ under the values w_i , it may be better to increase the set of active project classes again if, at the beginning of period 2, the values $\bar{w}_i$ turn out as the more accurate estimates, in order to finally reach concentration on a project class $\bar{i}_0$ that is more favorable than i_0 under the values $\bar{w}_i$.

4 Numerical solution

Since the emphasis of this article is on theoretical results, we only outline the question how to compute the optimal policy numerically by restricting ourselves to the important special cases of $T = 1$ and $T = 2$ periods. The case of only $T = 1$ period is easy, since, in view of Theorem 1, it can be treated by computing the objective function values of all extremal decisions $\mathbf{e}_i$ and choosing the best one. So let us turn to the case $T = 2$.

We use Theorem 4 which states that in period 2, the optimal decision is extremal, i.e., focused on a single project class i^* , and in period 1, it is focused on only two project classes. Consider a fixed decision $\mathbf{e}_{i^*}$ in period 2. Then, by (12)–(14),

$$\begin{aligned} G(\mathbf{x}_1, \mathbf{e}_{i^*}) &= G_1(\mathbf{x}_1) + G_2(\mathbf{x}_1, \mathbf{e}_{i^*}) \\ &= \sum_{i=1}^n w_i x_{i1} \varphi(z_{i1}) + \sum_{i=1}^n w_i \delta_{i,i^*} \varphi(z_{i1} - \beta_i + \eta_i x_{i1}) \\ &= \sum_{i \neq i^*} w_i x_{i1} \varphi(z_{i1}) + w_{i^*} x_{i^*1} \varphi(z_{i^*1}) + w_{i^*} \varphi(z_{i^*1} - \beta_{i^*} + \eta_{i^*} x_{i^*1}). \end{aligned} \quad (24)$$

We choose $i' \in \{1, \dots, n\} \setminus \{i^*\}$ in such a way that

$$w_{i'} \varphi(z_{i'1}) = \max \{w_i \varphi(z_{i1}) : i \in \{1, \dots, n\} \setminus \{i^*\}\}. \quad (25)$$

Then it can be seen from (24) that it is optimal in period 1 to concentrate all efforts on project classes i' and i^* (cf. also the proof of Theorem 4 in Appendix A). By this choice for $\mathbf{x}_1$, (24) reduces to

$$G(\mathbf{x}_1, \mathbf{e}_{i^*}) = w_{i'} (1 - x_{i^*1}) \varphi(z_{i'1}) + w_{i^*} x_{i^*1} \varphi(z_{i^*1}) + w_{i^*} \varphi(z_{i^*1} - \beta_{i^*} + \eta_{i^*} x_{i^*1}). \quad (26)$$

(Note that for $\mathbf{x}_1$ concentrating on i' and i^* , we have $x_{i'} + x_{i^*} = 1$.)

This gives the following algorithm for finding the optimal policy:

procedure optimize-two-stage-policy {
 set $G^{\text{best}} = 0$;
 for $i^* = 1$ to n {
 compute the optimal $i' \in \{1, \dots, n\} \setminus \{i^*\}$ according to (25);
 for given i' and i^* , determine the value of x_{i^*1} maximizing (26), e.g., by Newton's method;
 set $\mathbf{x}_1 = \mathbf{x}_1(i^*)$ to the decision investing effort $1 - x_{i^*1}$ into project class i' and effort x_{i^*1} into project class i^* ;
 if $(G(\mathbf{x}_1(i^*), \mathbf{e}_{i^*}) > G^{\text{best}})$ {
 set $G^{\text{best}} = G(\mathbf{x}_1(i^*), \mathbf{e}_{i^*})$;
 set $\mathbf{x}_1^{\text{best}} = \mathbf{x}_1(i^*)$;
 set $\mathbf{x}_2^{\text{best}} = \mathbf{e}_{i^*}$;
 }
 }
}

Example 5 Let φ be the logistic function defined by (1) with arbitrary a and b , and let $T = 2$. In this special case, the optimal policy can be computed by closed formulas. For fixed i^* , we abbreviate x_{i^*1} by u , and $G(\mathbf{x}_1, \mathbf{e}_{i^*})$ by $G(u)$. (Observe that i' is determined by i^* and hence also fixed.) A necessary condition for u being a maximum of

$$G(u) = w_{i'} (1 - u) \varphi(z_{i'1}) + w_{i^*} u \varphi(z_{i^*1}) + w_{i^*} \varphi(z_{i^*1} - \beta_{i^*} + \eta_{i^*} u)$$

is given by $G'(u) = 0$. One finds $G'(u) = C_1 + C_2 \varphi'(C_3 + C_4 u)$ with

$$C_1 = -w_{i'} \varphi(z_{i'1}) + w_{i^*} \varphi(z_{i^*1}), \quad C_2 = w_{i^*} \eta_{i^*}, \quad C_3 = z_{i^*1} - \beta_{i^*}, \quad C_4 = \eta_{i^*},$$

such that $G'(u) = 0$ amounts to $\varphi'(C_3 + C_4 u) = C$ with $C = -C_1/C_2$. Short calculation yields $\varphi'(z) = ab(1 + aw)^{-2} w$ with $w = \exp(-bz)$. Therefore, $\varphi'(z) = C$ if and only if $abw = C(1 + aw)^2$. This is a quadratic equation in w . From the solutions $w_{1,2}$, we obtain via $z = -b^{-1} \log w$ the corresponding values $z_{1,2}$, and from them, by means of $C_3 + C_4 u = z$, we get values $u_{1,2}$ as candidates for (local) maxima. By inserting into the second derivative $G''(u) = C_2 C_4 \varphi''(C_3 + C_4 u)$ of G , values that are not maxima can be sorted out. This yields to each fixed i^* the best assigned u .

Taking the maximum of the objective function values over $i^* = 1, \dots, n$ completes the determination of the optimal policy.

5 Generalized model

5.1 Project classes requiring several competencies

In the sequel, we extend the model of Sect. 2 to the case where a project class can require several different competencies, such that there is no one-to-one correspondence between project classes and competencies anymore. As a further generalization, we also admit the case where the supply of work time is not fixed, but can vary from period to period. We use again the index i for project classes, but introduce the new index r for competencies ($r = 1, \dots, R$). As in Gutjahr et al. (2008a,b), that part of work in project class i that requires competency r is denoted as *work package* (i, r) .

Projects require competencies to varying extent. We describe the extent to which project i requires competency i by the number d_{ir} . It is convenient to interpret the numbers d_{ir} as shares, i.e., we assume that they are normalized by the condition $\sum_{r=1}^R d_{ir} = 1 \forall i$. An investment of size y_i into project class i implies then that for work package (i, r) , effective work of an amount of $d_{ir} y_i$ has to be spent.

In the extended model, the index i has to be replaced in (2)–(8) by the index r wherever i rather refers to a competency than to a project class. In particular, we write now γ_{rt} and z_{rt} instead of γ_{it} and z_{it} , respectively, since efficiency values and competence scores are assigned to competencies rather than to project classes. Similarly, β_r and η_r is written instead of β_i and η_i , respectively. The set of variables x_{it} must be extended to the set of variables x_{irt} , where x_{irt} gives the amount of real work spent on work package (i, r) in period t .

By these extensions, we obtain the following generalized model:

$$\sum_{i=1}^n w_i y_i \rightarrow \max \quad (27)$$

$$\gamma_{rt} = \varphi(z_{rt}) \quad \forall r, t \quad (28)$$

$$z_{rt} = z_{r1} - \beta_r(t-1) + \eta_r \sum_{i=1}^n \sum_{s=1}^{t-1} x_{irs} \quad \forall r, t \quad (29)$$

$$\sum_{i=1}^n \sum_{r=1}^R x_{irt} = \sigma_t \quad \forall t \quad (30)$$

$$\sum_{t=1}^T \gamma_{rt} x_{irt} = d_{ir} y_i \quad \forall i, r \quad (31)$$

$$x_{irt} \geq 0 \quad \forall i, r, t \quad (32)$$

$$y_i \geq 0 \quad \forall i. \quad (33)$$

The objective function (27) is the same as that of (2). Equation (28), describing the relation between competence and efficiency, corresponds to Eq. (3). Equation (29)

extends Eq. (4); the essential difference is that the representation of the gain in competency r by learning needs now a summation over all project classes i . Equation (30) corresponds to Eq. (5), but again, a summation over all project classes i has to be performed in order to capture the overall real work time in period t . Furthermore, the overall available work time needs not to be unity, but is given by an amount σ_t that can depend on period t . In Eq. (31), the r.h.s. reads now $d_{ir}y_i$ instead of y_i as in (6), since only the effort connected with competency r is taken into account. Eqs. (32) and (33) are the nonnegativity constraints for the extended model.

It is easy to see that model (2)–(8) is obtained from model (27)–(33) as the special case $n = R$, $d_{ir} = \delta_{ir}$ and $\sigma_t = 1$: In this special case, Eq. (31) implies $\sum_{t=1}^T \gamma_{rt} x_{irt} = 0$ for $i \neq r$, whence $x_{irt} = 0$ for $i \neq r$ follows. As a consequence, Eq. (31) reduces to $\sum_{t=1}^T \gamma_{it} x_{iit} = y_i$ for all i , and Eq. (30) reduces to $\sum_{i=1}^n x_{iit} = 1$ for all t . Similarly, in Eq. (29), the sum $\sum_{i=1}^n \sum_{s=1}^{t-1} x_{irs}$ reduces to $\sum_{s=1}^{t-1} x_{iis}$. All that still has to be done to obtain the model of Sect. 2 is to abbreviate x_{iit} by x_{it} .

In the present form (27)–(33), the model would allow that the share d_{ir} of the effort y_i for project class i corresponding to competency r is spent at *any* time during the entire planning interval. In particular, this would mean that if work in project class i is started in period 1 and ended in, say, period 10, then it would be possible to invest the entire effort connected with competency 1 already in period 1, but to delay the effort connected with competency 2 to period 10. This may be not very realistic. For this reason, we modify our extended model by the additional constraint that the ratios between effective work times in the single competencies required by a project class remain constant over time. In formal terms, this constraint can be expressed as

$$\frac{\gamma_{rt} x_{irt}}{\gamma_{\rho t} x_{i\rho t}} = \frac{d_{ir}}{d_{i\rho}} \quad \forall i, r, \rho, t. \quad (34)$$

On the l.h.s. of (34), we have the ratio between effective work times in work packages (i, r) and (i, ρ) during period t . We require that this ratio does not change from period to period. Therefore, it must be identical to the corresponding ratio taken over the entire planning interval, which is $d_{ir}/d_{i\rho}$ because of (31).

The extended model (27)–(33) with the additional constraint (34) can be reduced to a more concise form in a similar way as (2)–(8) has been reduced to (9)–(10). From (34), one obtains $x_{i\rho t} = \gamma_{rt} x_{irt} d_{i\rho}/\gamma_{\rho t} d_{ir}$, and hence by summation over ρ ,

$$x_{it} = \left(\sum_{\rho=1}^R \frac{d_{i\rho}}{\gamma_{\rho t}} \right) \cdot \frac{\gamma_{rt} x_{irt}}{d_{ir}}, \quad (35)$$

where $x_{it} = \sum_{\rho=1}^R x_{i\rho t}$ denotes the overall real time invested into project class i in period t . Conversely, from (35) we get

$$x_{irt} = \frac{d_{ir}}{\gamma_{rt}} \left(\sum_{\rho=1}^R \frac{d_{i\rho}}{\gamma_{\rho t}} \right)^{-1} x_{it}. \quad (36)$$

Thus, x_{irt} is obtained by multiplying x_{it} by the normalized ratio d_{ir}/γ_{rt} between the effort d_{ir} for work package (i, r) and the efficiency γ_{rt} in competency r during period t . As a consequence,

$$\sum_{t=1}^T \gamma_{rt} x_{irt} = d_{ir} \sum_{t=1}^T \left(\sum_{\rho=1}^R \frac{d_{i\rho}}{\gamma_{\rho t}} \right)^{-1} x_{it},$$

and comparing this with (31), we find

$$y_i = \sum_{t=1}^T \left(\sum_{\rho=1}^R \frac{d_{i\rho}}{\gamma_{\rho t}} \right)^{-1} x_{it}. \quad (37)$$

Insertion of (37) into the objective function (27) yields

$$\sum_{i=1}^n w_i \sum_{t=1}^T \left(\sum_{r=1}^R \frac{d_{ir}}{\gamma_{rt}} \right)^{-1} x_{it} \rightarrow \max, \quad (38)$$

and insertion of (36) into (29) combined with (28) gives

$$\gamma_{rt} = \varphi \left(z_{r1} - \beta_r(t-1) + \eta_r \sum_{i=1}^n \sum_{s=1}^{t-1} \frac{d_{ir}}{\gamma_{rs}} \left(\sum_{\rho=1}^R \frac{d_{i\rho}}{\gamma_{\rho s}} \right)^{-1} x_{is} \right). \quad (39)$$

For x_{it} , the condition $\sum_{i=1}^n x_{it} = \sigma_t$ for all t has to be satisfied. Thus, (27)–(33) with the additional constraint (34) can be reformulated by (38)–(39) and the condition $\sigma_t^{-1} \mathbf{x}_t \in S_n$ for all t . By the substitution $\mathbf{x}_t = \sigma_t \bar{\mathbf{x}}_t$, we get $\bar{\mathbf{x}}_t \in S_n$, such that the feasible set is now again the simplex S_n . On the other hand, in (38) and (39), x_{it} and x_{is} have to be replaced by $\sigma_t \bar{x}_{it}$ and $\sigma_s \bar{x}_{is}$, respectively.

5.2 Solution of the generalized model

Concerning the numerical solution, we restrict ourselves again to the special cases $T = 1$ and $T = 2$. In the case $T = 1$, the problem takes the following form:

$$\sum_{i=1}^n w_i \left(\sum_{r=1}^R \frac{d_{ir}}{\varphi(z_{r1})} \right)^{-1} \sigma_1 \bar{x}_{i1} \rightarrow \max$$

subject to $\bar{\mathbf{x}}_1 \in S_n$. This is a linear program. It takes its solution in an extremal point $\mathbf{e}_{i^*}$ ($1 \leq i \leq n$) of the simplex S_n . Thus, in the case of only one period, it is optimal

to fully focus on one single project class i^* . Obviously,

$$i^* = \arg \max_{i=1, \dots, n} w_i \left(\sum_{r=1}^R \frac{d_{ir}}{\varphi(z_{r1})} \right)^{-1}.$$

Now let us turn to the special case $T = 2$ of two periods. For the sake of simplicity, let us assume $\sigma_1 = \sigma_2 = 1$, such that $\mathbf{x}_t \in S_n$. (The general case can be treated with only slight modifications.) From the consideration above concerning the case $T = 1$, we can immediately conclude that in period 2, an extremal decision (i.e., concentration on a single project class) is optimal. Of course, the optimal decision $\mathbf{e}_{i^*}$ in period 2 depends on the decision $\mathbf{x}_1$ made in period 1. From (38) and (39), we obtain the following problem:

$$\sum_{i=1}^n w_i \left(\sum_{r=1}^R \frac{d_{ir}}{\varphi(z_{r1})} \right)^{-1} x_{i1} + w_{i^*} \left(\sum_{r=1}^R \frac{d_{i^*r}}{\gamma_{r2}} \right)^{-1} \rightarrow \max \quad (40)$$

subject to

$$\gamma_{r2} = \varphi \left(z_{r1} - \beta_r + \eta_r \sum_{i=1}^n \frac{d_{ir}}{\varphi(z_{r1})} \left(\sum_{\rho=1}^R \frac{d_{i\rho}}{\varphi(z_{\rho1})} \right)^{-1} x_{i1} \right) \quad \forall r, \quad (41)$$

$$\mathbf{x}_1 \in S_n, \quad (42)$$

$$1 \leq i^* \leq n. \quad (43)$$

To each given $\mathbf{x}_1$, the second term in (40) can immediately be optimized by identifying that $i^* = 1, \dots, n$ for which $(1/w_{i^*}) \sum_{r=1}^R d_{i^*r}/\gamma_{r2}$ is minimal, where $\gamma_{r2} = \gamma_{r2}(\mathbf{x}_1)$ is computed from (41). In total, this gives a nonlinear maximization problem on the simplex S_n . In general, this problem is neither convex nor concave; multiple local optima may exist.

Example 6 Consider the following case:

$$w_1 = 100, \quad w_2 = 200, \quad (d_{ir}) = \begin{pmatrix} 0.85 & 0.15 \\ 0.25 & 0.75 \end{pmatrix}, \quad z_{11} = 2, \quad z_{21} = -1, \\ \beta_1 = 0.2, \quad \beta_2 = 0.4, \quad \eta_1 = 3, \quad \eta_2 = 0.5.$$

Figure 2 plots (40) as a function of x_{11} , where for each x_{11} , the optimal i^* has been inserted, and γ_{r2} has been computed from (41). We see that two local optima exist. Let us discuss this example from a managerial perspective. Evidently, it is not profitable in this case to invest exclusively into project class 1, although this class utilizes the

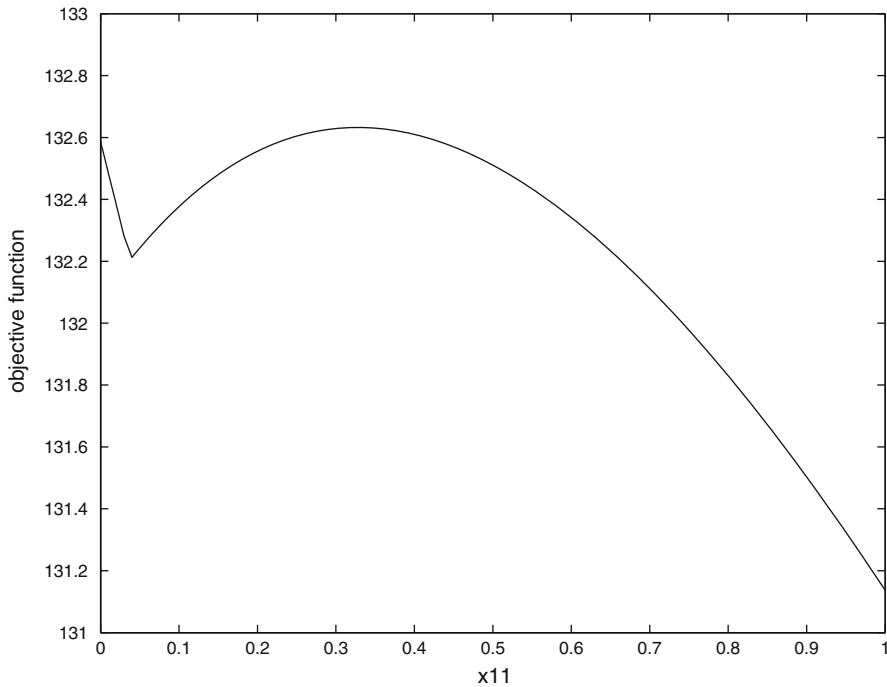


Fig. 2 Objective function in dependence of x_{11} in Example 6

leading competency of the firm (competency 1) to the largest extent: we can see that setting $x_{11} = 1$ entails the lowest possible return. Instead, in the considered situation, the firm has two almost equally good options⁴: (i) it can switch completely to project class 2, neglecting its leading competency 1 to a considerable degree. This choice corresponds to the left (marginal) local maximum at $x_{11} = 0$ in Fig. 2. (ii) Alternatively, the firm can also further expand its leading competency. As seen from the figure, this is optimally achieved by a portfolio split investing in the first period about a third of the available resources into project class 1 and two thirds into project class 2. This yields the local maximum at $x_{11} = 0.33$ in the figure (which is also the global maximum). In the case that this second option is chosen, a supposedly diversified portfolio investment stems from a focused *competence* investment. The example shows that hedging against uncertainty may not be the only reason for project portfolio diversification. Another reason can sometimes lie in the attempt to preserve a concentration in the “competence space”, even at the price of a diversification in the “project space”.

⁴ Which of these options the firm will choose in practice depends on the “heuristics” the management follows; Gino and Pisano (2006) have emphasized that in the R & D area, the portfolio decision making is often driven by heuristic rules.

5.3 Limited investments

There is another possible extension of our model that may be relevant for certain cases of applications: possibly, the amount of investment into a project class can be limited by an upper bound. E.g., this holds in a situation of limited demand. The model (27)–(33), with or without the additional constraint (34), can be extended in a straightforward way to capture also bounds of this type. All we have to do for this purpose is to add constraints of the form

$$y_i \leq c_i \quad \forall i. \quad (44)$$

Therein, c_i is an upper bound for the overall effective work time that can be invested into project class i . For simplicity, let us assume $\sigma_t = 1$ for all t in the sequel. In the version (38)–(39) where constraint (34) has been imposed and the variables y_i have been eliminated, (44) transforms to

$$\sum_{t=1}^T \left(\sum_{\rho=1}^R \frac{d_{i\rho}}{\gamma_{\rho t}} \right)^{-1} x_{it} \leq c_i \quad \forall i \quad (45)$$

in view of (37). In the case $T = 1$, the expressions $\left(\sum_{\rho=1}^R \frac{d_{i\rho}}{\gamma_{\rho t}} \right)^{-1}$ become constants (independent of the decision variables), such that the conditions (45) are linear and restrict the simplex S_n to a smaller convex polyhedron $\tilde{S}_n$. At least one extremal point of $\tilde{S}_n$ is optimal, which allows a fast solution. In the case $T = 2$, for fixed decision $\mathbf{x}_1$, the optimal decision in period 2 can be determined analogously as in the case $T = 1$, and considering this, $\mathbf{x}_1$ can be optimized essentially by the same techniques as the case of unlimited investments (see Sect. 5.2).

Example 7 Currently, the proposed model is being tested at a real-world application provided by the E-Commerce Competence Center Austria (EC3), a public–private partnership institution that is funded by the Austrian Federal Ministry of Economic Affairs and the City of Vienna as well as by several private companies. [Details concerning this application can be found in Gutjahr et al. (2008a,b).] Data describing competence scores in 80 different competencies (even at the level of individual employees) is available. In order to have a more comfortable granularity, the 80 single competencies were classified into eight groups, namely (a) data analysis and business analytics, (b) data science, (c) digital economy and society, (d) eBusiness components and foundations, (e) functional business domains, (f) symbol processing, (g) digital technologies, (h) economic activity categories, (i) methodological competence.

From the competence scores, initial efficiency values γ_{r1} were derived. In addition, we also determined the competence requirements for 18 different projects. The projects can be classified into six project classes, which enables an estimation of the values d_{ir} . Three different indicators of economic gains from each of the projects have been estimated by the management; the three numbers can be combined to a single value by a weighted average. (Since the organization is semi-private, not only economic profits matter.) In this way, the numbers w_i are obtained.

Part of the EC3 data refer to a 2-years planning horizon; it is already informative to investigate the two-stage decision process where a period corresponds to one year, which can be done by the algorithm indicated above.

In this application, upper bounds c_i on the investments into certain project classes *do* occur, e.g., in the case of a project class consisting of publicly funded projects. Although these projects have a relatively high rate of return on investment (where return is measured as indicated above), they cannot be expanded to an arbitrary degree. This leads to mixed portfolios also in the last period. Detailed results would exceed the scope of this paper and will be presented elsewhere.

6 Conclusions

We have proposed a model for optimizing the process of dynamic continuous project portfolio selection, taking account of changes in human competencies caused by learning on the one hand, forgetting and knowledge depreciation on the other hand. Let us outline some managerial insights that can be derived from the results.

Since, by learning, focused investments into projects that require certain competencies increase the ability of the firm to engage in projects of just this type in the future, one would expect that *specialization* is always a good strategy. Our theoretical results partially confirm this view: In the absence of uncertainty and of externally caused limits on investments into certain project classes, *and* in a basically static market environment, it is advisable indeed to let the current project portfolio converge to a single project class (the question to *which* class being nontrivial).

Nevertheless, our results say that usually, it is not optimal to *immediately* concentrate on a single project class. Rather than that, it seems that in order to keep the overall gains as high as possible, the focus on the core competence should be gradually “worked out” by a process of increasing specialization. An exception is a company that does not yet have considerable expertise in any of the project classes under consideration: in such a beginner’s situation, it is profitable to start with a highly specialized focus from the first moment.

For allowing practical implications, the picture provided by our deterministic model has still to be complemented by the effect of *uncertainty* (e.g., on future profits from the considered project classes) that disturbs the described ideal process of gradually increasing concentration in two different ways. First, *quantifiable* uncertainty constitutes what is called “decisions under risk” and is typically handled formally by means of a stochastic model. As it is well-known from a long-reaching tradition of works on resource allocation under risk, *diversification* is a possible mean of increasing expected returns without also increasing risk, or, vice versa, of reducing risk while keeping aspiration levels for expected returns. Evidently, in project portfolio management, diversification counteracts the process of gradual specialization proposed above. Thus, a risk-averse decision maker may choose to slow down the process of specialization towards a profitable core competency or even to reverse it for the sake of hedging against the risk implicit in assumptions that only estimate the future demand based on present information.

Second, there are also types of uncertainty that cannot be easily quantified by formal models—some (positive or negative) events occur unexpectedly, and their impact can nevertheless be very high. For the practice of project management, which is often done in a rolling-horizon fashion, this means that periodically or even spontaneously at some points in time, the parameters of the used model have to be adapted, and new opportunities have to be included into consideration. It is relatively clear what consequences this will have for the above-mentioned “ideal” process of gradually increasing concentration: the changed situation will divert the process from its current direction, such that temporarily, the project portfolio will become larger again. For this reason, it can be anticipated that usually, the final state of perfect concentration will never be reached. More probably, the typical situation will be characterized by a kind of equilibrium between diverging and converging trends.

In total, we arrive at the following picture: on the demand side, market risk prompts for diversification; on the supply side, however, the dynamic of competence accumulation prompts for gradual but steady specialization. Together, these two opposite forces create a non-negligible tension representing a continuing challenge to large organizations. It is this tension that also underlies Christensen’s observation about the innovator’s dilemma: Christensen (2003) shows that when firms succeeded, they usually did so because they invested aggressively in those technologies that were both well-grounded in available competencies and well-motivated by the customers’ needs. However, when firms subsequently *failed*, it was also because they followed the just-mentioned strategy! Although their achieved specialization was quite appropriate with respect to the development so far, they were finally taken by surprise by “disruptive” innovations that caused fundamental market changes.

In order to represent the effects of market uncertainty in a decision support approach, a generalization of our optimization model to the case where returns (and possibly also other parameters) are random variables would be of considerable interest. The resulting dynamic–stochastic optimization problems will certainly be challenging with respect to solution techniques. Presumably, scenario-tree approaches can be applied for this purpose.

In addition, for the deterministic special case, some interesting questions are still open. To give some examples: are there generalizations of Theorems 1–4 to the extended model for project classes requiring several competencies? What happens if the presented model is modified by *discounting* returns from projects (this would even allow to consider the case of an infinite planning horizon)? What effects have bounds for investments on the properties of optimal policies? No doubt, a larger body of analytical results on such questions would contribute to our understanding of the core dynamics of project portfolio management taking competence evolution into account, and it would possibly enable the development of decision support systems that considerably facilitate managerial decisions in this area. On the way to the last goal, also a good deal of empirical research is still required.

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Appendix A

Proof of Theorem 1 Let us define

$$\begin{aligned}\bar{G}(\mathbf{x}) &= \sum_{i=1}^n x_i G(\mathbf{e}_i) = \sum_{i=1}^n x_i \sum_{j=1}^n w_j \delta_{ij} \sum_{t=1}^T \varphi(z_{j1} + (\eta_j \delta_{ij} - \beta_j)(t-1)) \\ &= \sum_{i=1}^n x_i w_i \sum_{t=1}^T \varphi(z_{i1} + (\eta_i - \beta_i)(t-1)).\end{aligned}$$

Since $\eta_i(t-1) \geq \eta_i x_i(t-1)$ for all i because of $0 \leq x_i \leq 1$, and since φ is nondecreasing, it follows that $\bar{G}(\mathbf{x}) \geq G(\mathbf{x})$ for all $\mathbf{x}$. The function $\bar{G}(\mathbf{x})$ is linear in $\mathbf{x} \in S_n$, hence it takes its maximum in one of the extremal points of the convex set S_n , say in $\mathbf{e}_{i^*}$. In the point $\mathbf{e}_{i^*}$, the functions $\bar{G}$ and G coincide: $\bar{G}(\mathbf{e}_{i^*}) = G(\mathbf{e}_{i^*})$. Therefore, for any $\mathbf{x} \in S_n$, we have

$$G(\mathbf{x}) \leq \bar{G}(\mathbf{x}) \leq \max_{1 \leq i \leq n} \bar{G}(\mathbf{e}_i) = \bar{G}(\mathbf{e}_{i^*}) = G(\mathbf{e}_{i^*}).$$

This implies that $\mathbf{e}_{i^*}$ also maximizes G . □

Proof of Theorem 2 The proof uses induction with respect to T . For $T = 1$ (case of only one period), we have the scenario of investments that remain constant over time, hence the assertion is true by Theorem 1. Now let us fix some $T > 1$ and assume that the assertion holds for the case of $T - 1$ periods. By (12)–(14), the function $G(X)$ for T periods can be decomposed as follows:

$$G(X) = G_1(X) + \sum_{t=2}^T G_t(X) = \sum_{i=1}^n w_i x_{i1} \varphi(z_{i1}) + \sum_{t=2}^T G_t(X). \quad (46)$$

Let us first look at the second term of this decomposition, considering the variables x_{i1} occurring in the first term as fixed. Remind that $G_t(X)$ only depends on $\mathbf{x}_1, \dots, \mathbf{x}_t$. For fixed chosen $\mathbf{x}_1 = (x_{11}, \dots, x_{n1})^T$ and $t \geq 2$,

$$\begin{aligned}G_t(X) &= \sum_{i=1}^n w_i x_{it} \varphi\left(z_{i1} - \beta_i - \beta_i(t-2) + \eta_i x_{i1} + \eta_i \sum_{s=2}^{t-1} x_{is}\right) \\ &= \sum_{i=1}^n w_i x_{it} \varphi\left(\tilde{z}_{i1} - \beta_i(t-2) + \eta_i \sum_{s=2}^{t-1} x_{is}\right)\end{aligned} \quad (47)$$

with $\tilde{z}_{i1} = z_{i1} - \beta_i + \eta_i x_{i1}$. Setting $\tilde{t} = t - 1$ and $\tilde{x}_{is} = x_{is,s+1}$ and using Eq. (47), we obtain

$$\begin{aligned}
\sum_{t=2}^T G_t(X) &= \sum_{\tilde{t}=1}^{T-1} G_{\tilde{t}+1}(X) = \sum_{\tilde{t}=1}^{T-1} \sum_{i=1}^n w_i x_{i,\tilde{t}+1} \varphi \left(\tilde{z}_{i1} - \beta_i(\tilde{t} - 1) + \eta_i \sum_{s=2}^{\tilde{t}} x_{is} \right) \\
&= \sum_{\tilde{t}=1}^{T-1} \sum_{i=1}^n w_i \tilde{x}_{i\tilde{t}} \varphi \left(\tilde{z}_{i1} - \beta_i(\tilde{t} - 1) + \eta_i \sum_{s=1}^{\tilde{t}-1} \tilde{x}_{is} \right) \\
&= \sum_{i=1}^n w_i \sum_{\tilde{t}=1}^{T-1} \tilde{x}_{i\tilde{t}} \varphi \left(\tilde{z}_{i1} - \beta_i(\tilde{t} - 1) + \eta_i \sum_{s=1}^{\tilde{t}-1} \tilde{x}_{is} \right), \tag{48}
\end{aligned}$$

and this is just the objective function (9) for the reduced problem with $T - 1$ periods and initial values $\tilde{z}_{i1}$ of the competence scores, expressed in the decision variables $\tilde{x}_{i\tilde{t}}$. By the induction assumption, the last problem has a solution $\tilde{X}^*$ that consists only of extremal points of S_n . Therefore, to each $\mathbf{x}_1 \in S_n$, there is a sequence $i_2(\mathbf{x}_1), \dots, i_T(\mathbf{x}_1)$ of indices (depending on $\mathbf{x}_1$) such that an optimal solution $(\mathbf{x}_2, \dots, \mathbf{x}_n)$ of the problem

$$\sum_{t=2}^T G_t(X) = \sum_{t=2}^T G_t(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_t) \rightarrow \max \tag{49}$$

for fixed given $\mathbf{x}_1$ is provided by $\mathbf{x}_t = \mathbf{e}_{i_s(\mathbf{x}_1)}$ ($s = 2, \dots, t$). It should be observed that problem (49) depends on $\mathbf{x}_1$ only via the values $\tilde{z}_{i1}$. Since we know now that for fixed $\mathbf{x}_1$, (49) takes its solution in a sequence of extremal points, we can write the optimal objective function value of (49) as

$$\max \left\{ \sum_{t=2}^T G_t(\mathbf{x}_1, \mathbf{e}_{i_2}, \dots, \mathbf{e}_{i_T}) : i_k \in \{1, \dots, n\} \ (\kappa = 2, \dots, T) \right\}. \tag{50}$$

Now let us consider the influence of $\mathbf{x}_1$. By definition of G_t ,

$$\sum_{t=2}^T G_t(\mathbf{x}_1, \mathbf{e}_{i_2}, \dots, \mathbf{e}_{i_T}) = \sum_{t=2}^T \sum_{i=1}^n w_i \delta_{i,i_t} \varphi \left(z_{i1} - \beta_i(t - 1) + \eta_i x_{i1} + \eta_i \sum_{s=2}^{t-1} \delta_{i,i_s} \right).$$

For fixed $(i_2, \dots, i_T)$, this is a convex function of the variables x_{i1} ($i = 1, \dots, n$) because of the convexity of φ . Since the maximum of convex functions is convex again, also (50) is a convex function of the variables x_{i1} ($i = 1, \dots, n$). Furthermore, $G_1(\mathbf{x}_1) = \sum_{i=1}^n w_i x_{i1} \varphi(z_{i1})$ is linear and hence also convex in the variables x_{i1} ($i = 1, \dots, n$). Therefore, in total, the function

$$F(\mathbf{x}_1) = G_1(\mathbf{x}_1) + \max \left\{ \sum_{t=2}^T G_t(\mathbf{x}_1, \mathbf{e}_{i_2}, \dots, \mathbf{e}_{i_T}) : i_k \in \{1, \dots, n\} \ (\kappa = 2, \dots, T) \right\}, \tag{51}$$

whose maximum is identical to the maximum of $G(X)$, is a convex function of $\mathbf{x}_1$ as well. This implies that also $\mathbf{x}_1$ takes its maximum in an extremal point of S_n . $\square$

Proof of Theorem 3 For abbreviation, we write $G(i_1, \dots, i_T)$ instead of $G(\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_T})$. Setting $x_{it} = \delta_{i, i_t}$ in (13), one obtains in our scenario where all β_i vanish that

$$\begin{aligned} G(i_1, \dots, i_T) &= \sum_{t=1}^T \sum_{i=1}^n w_i x_{it} \varphi \left(z_{i1} + \eta_i \sum_{s=1}^{t-1} x_{is} \right) \\ &= \sum_{t=1}^T \sum_{i=1}^n w_i \delta_{i, i_t} \varphi \left(z_{i1} + \eta_i \sum_{s=1}^{t-1} \delta_{i, i_s} \right) \\ &= \sum_{t=1}^T w_{i_t} \varphi \left(z_{i_t, 1} + \eta_{i_t} \sum_{s=1}^{t-1} \delta_{i_t, i_s} \right). \end{aligned}$$

Therein, $\sum_{s=1}^{t-1} \delta_{i_t, i_s}$ is the number of indices $s \in \{1, \dots, t-1\}$ for which $i_s = i_t$. Moreover, in analogy to definition (12), we set

$$G_t(i_1, \dots, i_T) = G_t(i_1, \dots, i_t) = w_{i_t} \varphi \left(z_{i_t, 1} + \eta_{i_t} \sum_{s=1}^{t-1} \delta_{i_t, i_s} \right),$$

such that

$$G(i_1, \dots, i_T) = G_1(i_1) + G_2(i_1, i_2) + \dots + G_T(i_1, \dots, i_T).$$

(Note that $G_t(i_1, \dots, i_T)$ actually depends only on $i_1, \dots, i_t$.)

We show the following auxiliary statement: If for a policy $(i_1, \dots, i_T)$, the last $T - \tau + 1$ entries ($\tau \in \{2, \dots, T\}$) are identical, i.e., if

$$(i_1, \dots, i_T) = (i_1, \dots, i_{\tau-1}, i_\tau, \dots, i_\tau) \quad (2 \leq \tau \leq T),$$

where on the r.h.s., the element i_τ occurs $T - \tau + 1$ times, then $(i_1, \dots, i_T)$ can only be optimal, if also $i_{\tau-1} = i_\tau$. The truth of this statement implies the assertion of the theorem by straightforward induction.

An indirect proof is used to verify the auxiliary statement. Assume that

$$x = (i_1, \dots, i_{\tau-2}, i_{\tau-1}, i_\tau, i_\tau, \dots, i_\tau)$$

is optimal with $i_{\tau-1} \neq i_\tau$. We modify x on position τ by setting

$$x^+ = (i_1, \dots, i_{\tau-2}, i_{\tau-1}, i_{\tau-1}, i_\tau, \dots, i_\tau),$$

and we modify x on position $\tau - 1$ by setting

$$x^- = (i_1, \dots, i_{\tau-2}, i_\tau, i_\tau, i_\tau, \dots, i_\tau).$$

Let us introduce the following abbreviations:

$$w = w_{i_{\tau-1}}, \quad w' = w_{i_{\tau}}, \quad \eta = \eta_{i_{\tau-1}}, \quad \eta' = \eta_{i_{\tau}}, \quad z = z_{i_{\tau-1}, \tau-1}, \quad z' = z_{i_{\tau}, \tau-1}.$$

Observe that the competence scores z and z' in period $\tau - 1$ corresponding to competencies $i_{\tau-1}$ and i_{τ} , respectively, only depend on $i_1, \dots, i_{\tau-2}$, such that they are identical for the three policies x, x^+ and x^- . Furthermore, let

$$C = \sum_{t=1}^{\tau-2} G_t(x) = \sum_{t=1}^{\tau-2} G_t(x^+) = \sum_{t=1}^{\tau-2} G_t(x^-).$$

One obtains

$$\begin{aligned} G(x) &= C + w \varphi(z) + w' \varphi(z') + w' \sum_{t=\tau+1}^T \varphi(z' + \eta'(t - \tau)), \\ G(x^+) &= C + w \varphi(z) + w \varphi(z + \eta) + w' \sum_{t=\tau+1}^T \varphi(z' + \eta'(t - \tau - 1)), \end{aligned}$$

and

$$G(x^-) = C + w' \varphi(z') + w' \varphi(z' + \eta') + w' \sum_{t=\tau+1}^T \varphi(z' + \eta'(t - \tau + 1)).$$

The first and the third equation can be re-written as

$$G(x) = C + w \varphi(z) + w' \sum_{t=\tau+1}^T \varphi(z' + \eta'(t - \tau - 1)) + w' \varphi(z' + \eta'(T - \tau))$$

and

$$\begin{aligned} G(x^-) &= C + w' \sum_{t=\tau+1}^T \varphi(z' + \eta'(t - \tau - 1)) \\ &\quad + w' \varphi(z' + \eta'(T - \tau)) + w' \varphi(z' + \eta'(T - \tau + 1)), \end{aligned}$$

respectively. Since x is optimal by assumption, $G(x^+) \leq G(x)$, i.e.,

$$w \varphi(z + \eta) \leq w' \varphi(z' + \eta'(T - \tau)).$$

Since $\eta > 0$, $w > 0$, and φ is strictly increasing, it follows that

$$\begin{aligned} G(x^-) &\geq C + w' \sum_{t=\tau+1}^T \varphi(z' + \eta'(t-\tau-1)) + w \varphi(z + \eta) + w' \varphi(z' + \eta'(T-\tau+1)) \\ &> C + w' \sum_{t=\tau+1}^T \varphi(z' + \eta'(t-\tau-1)) + w \varphi(z) + w' \varphi(z' + \eta'(T-\tau)) = G(x), \end{aligned}$$

in contradiction to the assumption that x is optimal. This completes the proof. $\square$

Proof of Theorem 4 We use (14) and (12) and apply induction with respect to T . For $T = 1$,

$$G(X) = G_1(\mathbf{x}_1) = \sum_{i=1}^n w_i x_{i1} \varphi(z_{i1}),$$

and because of the linearity of this objective function, there is an optimal decision $\mathbf{x}_1$ which is an extremal point of S_n , i.e., $\mathbf{x}_1 = \mathbf{e}_i$ for some $i \in \{1, \dots, n\}$. Since $|\text{support}(\mathbf{e}_i)| = 1$, this verifies the assertion.

Now let $T > 1$, and assume that the assertion is true for the case of $T - 1$ periods. By (46) and (48), we have $G(X) = G_1(X) + \sum_{t=2}^T G_t(X)$, where

$$\sum_{t=2}^T G_t(X) = \sum_{\tilde{t}=1}^{T-1} \sum_{i=1}^n w_i \tilde{x}_{i\tilde{t}} \varphi \left(\tilde{z}_{i1} - \beta_i(\tilde{t}-1) + \eta_i \sum_{s=1}^{\tilde{t}-1} \tilde{x}_{is} \right) \quad (52)$$

with $\tilde{t} = t - 1$, $\tilde{\mathbf{x}}_s = \mathbf{x}_{s+1}$ and $\tilde{z}_{i1} = z_{i1} - \beta_i + \eta_i x_{i1}$. As in the proof of Theorem 2, we conclude that (52) is the objective function of the problem with $T - 1$ periods and initial values $\tilde{z}_{i1}$, expressed in the decision variables $\tilde{x}_{i\tilde{t}}$. By induction assumption, this problem has a solution $\tilde{X}^* = (\tilde{\mathbf{x}}_1^*, \dots, \tilde{\mathbf{x}}_{T-1}^*)$ with

$$|\text{support}_{\tilde{t}}^+(\tilde{X}^*) \setminus \text{support}_{\tilde{t}+1}^+(\tilde{X}^*)| \leq 1 \quad (1 \leq \tilde{t} \leq T-2).$$

Re-writing the last equation in terms of $t = \tilde{t} + 1$ and $\mathbf{x}_t = \tilde{\mathbf{x}}_{t-1}$ yields

$$|\text{support}_t^+(X^*) \setminus \text{support}_{t+1}^+(X^*)| \leq 1 \quad (2 \leq t \leq T-1), \quad (53)$$

which is (22) except for the case $t = 1$. Via the values $\tilde{z}_{i1}$, the solution $\tilde{X}^*$ depends on $\mathbf{x}_1$, so that more precisely, $\tilde{X}^* = \tilde{X}^*(\mathbf{x}_1)$, and analogously for its components: $\tilde{\mathbf{x}}_t^* = \tilde{\mathbf{x}}_t^*(\mathbf{x}_1)$ ($\tilde{t} = 1, \dots, T-1$).

Now we consider the decision $\mathbf{x}_1 \in S_n$ in period 1 for which the corresponding policy for the remaining periods,

$$\tilde{X}^*(\mathbf{x}_1) = (\tilde{\mathbf{x}}_1^*(\mathbf{x}_1), \dots, \tilde{\mathbf{x}}_{T-1}^*(\mathbf{x}_1)) = (\mathbf{x}_2^*(\mathbf{x}_1), \dots, \mathbf{x}_T^*(\mathbf{x}_1)),$$

produces together with $\mathbf{x}_1$ a maximal value of the objective function G . Let us denote this optimal decision $\mathbf{x}_1$ by $\mathbf{x}_1^*$. In the sequel, $\tilde{X}^*(\mathbf{x}_1^*)$ will be abbreviated by $\tilde{X}^*$, and $\mathbf{x}_t^*(\mathbf{x}_1^*)$ by $\mathbf{x}_t^*$ ($t = 2, \dots, T$). Evidently, $X^* = (\mathbf{x}_1^*, \mathbf{x}_2^*, \dots, \mathbf{x}_T^*)$ is then an optimal solution to the given problem.

With $\mathcal{T}_2 = \text{support}_2^+(X^*)$, we have

$$\begin{aligned} G(X^*) &= \sum_{i \notin \mathcal{T}_2} w_i x_{i1}^* \varphi(z_{i1}) + \sum_{i \in \mathcal{T}_2} w_i x_{i1}^* \varphi(z_{i1}) \\ &\quad + \sum_{t=2}^T \sum_{i=1}^n w_i x_{it}^* \varphi \left(z_{i1} - \beta_i(t-1) + \eta_i \sum_{s=1}^{t-1} x_{is}^* \right). \end{aligned} \quad (54)$$

In the double sum in (54), only those x_{it}^* ($t \geq 2$) can be different from zero for which $i \in \text{support}(\mathbf{x}_1^*) \subseteq \mathcal{T}_2$.

Now we can distinguish the two cases (i) $x_{i1}^* = 0$ for all $i \notin \mathcal{T}_2$, and (ii) $x_{i1}^* > 0$ for some $i \notin \mathcal{T}_2$. In case (i), we have $\text{support}(\mathbf{x}_1^*) \subseteq \mathcal{T}_2$ and therefore $|\text{support}_1^+(X^*) \setminus \mathcal{T}_2| = 0$, such that X^* satisfies (22) also for $t = 1$. In case (ii), let us choose $i' \in \{1, \dots, n\} \setminus \mathcal{T}_2$ in such a way that

$$w_{i'} \varphi(z_{i'1}) = \max \{w_i \varphi(z_{i1}) : i \in \{1, \dots, n\} \setminus \mathcal{T}_2\},$$

and set $\bar{X}^* = (\bar{\mathbf{x}}_1^*, \mathbf{x}_2^*, \dots, \mathbf{x}_T^*)$ with $\bar{\mathbf{x}}_1^*$ defined by

$$\bar{x}_{i1}^* = x_{i1}^* \quad \text{for all } i \in \mathcal{T}_2, \quad \bar{x}_{i'1}^* = 1 - \sum_{i \in \mathcal{T}_2} x_{i1}^*, \quad \text{and} \quad \bar{x}_{i1}^* = 0 \quad \text{for all } i \notin \mathcal{T}_2 \cup \{i'\}.$$

In other words, we redirect all efforts spent in period 1 on project classes not contained in $\mathcal{T}_2$ to a single project class not contained in $\mathcal{T}_2$ (providing maximum gain), such that the sum over $i \notin \mathcal{T}_2$ in (54) contracts to a single term. Then it is easy to see that $G_1(\bar{\mathbf{x}}_1^*) \geq G_1(\mathbf{x}_1^*)$. The double sum in (54), however, remains unchanged if X^* is replaced by $\bar{X}^*$, since terms x_{it}^* occur only in the sum $\sum_{s=1}^{t-1} x_{is}^*$, but this sum is multiplied by a non-vanishing factor x_{it}^* only if $i \in \mathcal{T}_2$.

Therefore, by a change from X^* to $\bar{X}^*$, the objective function value cannot decrease. Hence also $\bar{X}^*$ is optimal. The components $\mathbf{x}_2^*, \dots, \mathbf{x}_T^*$ satisfy condition (22) for $t \geq 2$ because of (53). Finally, for the component $\bar{\mathbf{x}}_1^*$, it holds that $\text{support}(\bar{\mathbf{x}}_1^*) \subseteq \mathcal{T}_2 \cup \{i'\}$, and thus $\bar{X}^*$ satisfies (22) also for $t = 1$.

It remains to show that also (23) holds. The lower bound is trivial. The upper bound follows by induction with respect to t from (22): it has already been shown that $|\text{support}(\mathbf{x}_T^*)| = 1$, so (23) is valid for $t = T$. Assume that (23) holds for some $t \leq T$. Then (22) implies

$$|\text{support}_{t-1}^+(X^*)| \leq |\text{support}_t^+(X^*)| + 1 \leq (T - t + 1) + 1 = T - (t - 1) + 1,$$

which verifies (23) for $t - 1$. Thus, everything is shown. $\square$

Appendix B

Derivation of the function φ from the Argote–Epplé–Darr model

When comparing the model of Sect. 2 to that in Argote and Epplé (1990) and Darr et al. (1995), three differences have to be kept in mind: first, our model is not based on discrete production units, but allows continuous (fractional) funding. This suggests an asymptotic consideration where the size of the production units tends to zero. Secondly, in our model, the efficiency $\varphi(z)$ has an upper bound, scaled to the value 1 by definition. In contrast to that, the required effort y as specified by the learning curve (20) converges to zero as $x \rightarrow \infty$, which means that after a sufficiently long continuation of production and in absence of forgetting, the efficiency can exceed any limit. However, the last-mentioned difference is not essential as long as we restrict ourselves to a fixed finite time interval. (We omit details for the sake of brevity.) Third, in our model, we do not assume that efficiency gains or losses occurring during a period already act during this same period, but rather that they become efficacious at the end of the period. Obviously, this approximation is not fully consistent with the classical learning curve model, neither in its discrete form nor in a “continuous” form (which will be described below) where the size of the units becomes infinitesimal. Therefore, we do not claim to be able to translate one of the models into the other. Instead, we are only interested in investigating which shape the learning function φ , as introduced in Sect. 2, would show in a continuous modification of a model of the type as presented in Argote and Epplé (1990) and Darr et al. (1995).

We start with the case where only learning takes place, and discuss forgetting later. Concerning the parameter b in the learning curve, we assume $0 < b < 1$.

In order to construct a continuous learning scheme, we derive from (20) a sequence of scaled learning curves of the form $y_{(p)} = a_{(p)} x_{(p)}^{-b}$ ($p = 1, 2, \dots$) with

$$a_{(p)} = p^{b-1} a_{(1)}, \quad x_{(p)} = p x_{(1)} \quad (p = 1, 2, \dots),$$

where we imagine that the p th process is obtained from the original process by splitting each unit into p parts. The scaling factor $a_{(p)}$ is chosen in such a way that after the production of $x_{(p)}$ (split) units, for each p , the same efficiency results from the learning effect (see below).

Interpreting the production effort as the *real* work time and the number of produced units as proportional to *effective* work time, we assume that each unit in the p th process contributes to the effective work time by the same amount $c_{(p)}$. In view of the splitting construction, we set $c_{(p)} = (1/p) c_{(1)}$. Then, the efficiency for the production of unit $x_{(p)}$ computes as

$$\begin{aligned} \gamma &= \frac{\text{effective work time}}{\text{real work time}} = \frac{c_{(p)}}{a_{(p)} x_{(p)}^{-b}} = c_{(p)} a_{(p)}^{-1} x_{(p)}^b \\ &= (1/p) c_{(1)} p^{1-b} a_{(1)}^{-1} p^b x_{(1)}^b = c_{(1)} a_{(1)}^{-1} x_{(1)}^b, \end{aligned}$$

which is independent of p .

Now, we let $p \rightarrow \infty$, such that a continuous process results. Let u denote the (cumulative) *real* work time invested until a certain moment, and let $\xi(u)$ be the corresponding (cumulative) *effective* work time up to this moment. By $\gamma(\xi)$, we denote the current efficiency achieved after ξ units of effective work time. Having already invested u units of real work time, after spending an additional infinitesimal amount du of real work time, the cumulative effective work time increases to $\xi(u + du) = \xi(u) + \gamma(\xi(u)) du$, whence we obtain the differential equation

$$\frac{d\xi(u)}{du} = \gamma(\xi(u)). \quad (55)$$

Since each unit in the p th process contributes to the effective work time by $c_{(p)}$, the effective work time after the production of $x_{(p)}$ units is $\xi = c_{(p)} x_{(p)}$, which is equal to $c_{(1)} x_{(1)}$ and hence independent of p . (Note that in the limit $p \rightarrow \infty$, the distinction between “efficiency during the production of a unit” and “efficiency immediately after the production of a unit” vanishes.) This gives $\gamma(\xi) = c_{(p)} a_{(p)}^{-1} x_{(p)}^b = c_{(p)} a_{(p)}^{-1} c_{(p)}^{-b} \xi^b = c_{(p)}^{1-b} a_{(p)}^{-1} \xi^b = c_{(1)}^{1-b} a_{(1)}^{-1} \xi^b$. Setting $C_1 = c_{(1)}^{1-b} a_{(1)}^{-1}$, we obtain $\gamma(\xi) = C_1 \xi^b$. Inserted into (55), this yields the differential equation $d\xi/du = C_1 \xi^b$, which can easily be solved: Using the initial condition $\xi(0) = 0$, we obtain $\xi = C_2 u^{1/(1-b)}$ with $C_2 = [(1-b)C_1]^{1/(1-b)}$. Insertion into $\gamma(\xi) = C_1 \xi^b$ yields

$$\gamma(\xi(u)) = C u^\alpha \quad (56)$$

with $C = C_1 C_2^b$ and $\alpha = b/(b-1)$. Compare now (56) with $\gamma = \varphi(z_0 + \eta u)$, which is the special case $\beta = 0$ of our basic relation $\gamma = \varphi(z_0 - \beta + \eta u)$ deriving efficiencies from real work times. We see that this relation can be obtained by using the specific learning function $\varphi(z) = z^\alpha$ analyzed in Example 4 and setting $z_0 = 0$ and $\eta = C^{1/\alpha}$. After short calculation, one finds $\eta = (1-b) c_{(p)}^{(1-b)/b} a_{(p)}^{-1/b}$, which is actually independent of p .

For $0 < b < 1/2$, $b = 1/2$ and $1/2 < b < 1$, we obtain the cases of a concave, linear and convex learning function φ , respectively.

Finally, let us turn to the case of forgetting or knowledge depreciation. In Darr et al. (1995), a model is used that can be shortly summarized as follows: the “accumulated stock of learning” at time t is not derived from the cumulative production until time t , but rather from a quantity K_t that is related to the number q_t of units produced in time interval $[t-1, t]$ by the discounting law $K_t = \lambda K_{t-1} + q_t$. The boundary case $\lambda = 1$ represents the case without forgetting. On the other hand, if $\lambda < 1$, knowledge depreciates over time. Consider the special case $q_t = 1$ for all t where the production rate is constant. Then, the number of produced units until time t is $x = t$, such that we can replace t with x . Unfolding the recursion, we get

$$K_x = \lambda^{x-1} + \lambda^{x-2} + \dots + \lambda + 1 = \frac{\lambda^x - 1}{\lambda - 1}.$$

Thus, the derivation above for the case without forgetting would have to be modified by replacing $y = ax^{-b}$ with $y = a(\frac{\lambda^x - 1}{\lambda - 1})^{-b}$. It seems evident that an analysis of a

corresponding continuous process along the lines of the derivations above does not lead anymore to a closed-form solution as in the case without forgetting.

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